



SPRING MID SEMESTER EXAMINATION-2025

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Subject Name: Engineering Economics
[Subject Code: HS30101]

Time: 1 1/2 Hours

Full Mark: 20

Answer Any four questions including question No.1 which is compulsory.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 Mark X 5]
 - a) If you deposit \$4000 into an account paying 6% annual interest compounded quarterly, how much money will be in the account after 5 years?
 - b) A certain sum of money becomes three times in 20 years at simple interest. In how many years does it become double itself at the same rate of simple interest?
 - c) An investor deposited Rs. 1,000 in a scheme for 2 years. The scheme offers 10% interest with quarterly compounding. Find the maturity value of the scheme.
 - d) What is the difference between substitutes and complements?
 - e) What is the difference between Giffen goods and inferior goods?
2.
 - a) Explain the consumer's equilibrium with the help of indifference curves and budget line. [3+2]
 - b) Jack needed a \$1191 in 3 years to be off some debt. How much should Jack put in a saving bank account that earns 6% today? [3+2]
3.
 - a) Diagrammatically show the followings:
 - (i) Increase in Demand is less than decrease in supply
 - (ii) Decrease in Demand is more than increase in supply.
 - (iii) Increase in Demand is more than increase in supply.
 - b) Price elasticity of demand of a commodity is (-) 2. A consumer demands 50 units of this when its price is 10 per unit. At what price he will demand 40 units of this commodity?

[3+2]

4. a) The following table relates to the tourist arrivals from 1990 to 1996 in India:

Years	1990	1991	1992	1993	1994	1995	1996
Tourist arrivals (in millions)	18	20	23	25	24	28	30

Fit a straight line trend by the method of least squares and estimate the number of tourists that would arrive in the year 2000.

- b) The quantity demanded of a commodity increases from 8,000 units to 10,000 units due to increase in income from Rs 60,000 to Rs 1,20,000. Find the income elasticity of demand.

[3+2]

5. a) Calculate TR, AR and MR using the following information below.

Price per unit (Rs)	Quantity Demanded (units)
1	100
2	90
3	70
4	60
5	50
6	40

- b) Madhu wants to purchase a home worth Rs 75,00,000. She will pay emi for the next 20 years towards her loan repayment. The rate of interest charged by the bank is 15% compounded annually. Find the amount of emi she will pay every year.

*** Best of Luck ***